1 Question

Write a program that prints terms of the following mathematical sequence:

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \dots$$

Your program should ask user to enter a double number representing a maximum limit, and should repeatedly add and print terms of the sequence until the overall sum of terms meets or exceeds the maximum limit that was taken from the user.

```
Enter the limit: 2
1 + 1/2 + 1/3 + 1/4 = 2.083
```

2 Question

Write a function that gets two integers M and N as parameters and returns the least common multiple of M and N.

```
LCM(12, 20) = 60
```

3 Question

Write a program that gets an integer number from the user and returns the number of parity (even or odd) switches between the digits (that is, the number of pairs of consecutive digits where one is even and the other is odd, in either order). For example, if user enters 36487, the program prints 2 because the parity switches between the 3 and the 6 and again between the 8 and the 7. If the number contains only one digit, or only even or odd digits, the program prints 0.

```
Enter a number: 800
Count of Parity Switches: 0

Enter a number: 12345
Count of Parity Switches: 4
```

4 Question

Write a program that asks user to enter an odd integer M. The program then prints (M + 1) / 2 lines where (i) each line contains M characters; (ii) N'th line has a $N-1$ '*' characters in the beginning and in the end; (iii) the remaining characters are "+".

```
Enter size: 5
+++++
*+++*
*****

Enter size: 7
+++++++
*+++++*
**+++**
*******
```

5 Question

Let say you have the function **dice**

```
dice()
```

which randomly throws a dice and returns the result. DO NOT WRITE THIS FUNCTION.

Write the first function **pair**

```
pair(int dice1, int dice2, int dice3)
```

which returns true if there are at least two dices whose values are equal, false otherwise.

Write the second function **triple**

```
triple(int dice1, int dice2, int dice3)
```

which returns true if all three dices are equal, false otherwise.

Write the third function **diceGame**

```
diceGame()
```

which plays the following dice game using the above functions. There are two players. The first player throws three dices. The second player throws three dices. Then the dices of players are compared according to this rule: A triple in three dices is better than a pair in three dices which is better than nothing.

```
First player: 1 2 1
Second player: 5 5 5
Second player wins

First player: 1 5 6
Second player: 4 1 2
No one wins
```

6 Question

Let say you have the function **isPrime**

```
public static bool isPrime(int N)
```

which returns true is N is prime, false otherwise. DO NOT WRITE THIS FUNCTION.

Write the first function **largestPower**

```
public static int largestPower(int N, int x)
```

which returns the largest power of x, which divides N. For example, degree(5040, 2) returns 4, because 2^4 is the largest power of 2 which divides 5040. Another example, degree(120, 3) returns 1, because 3^1 is the largest power of 3 which divides 120.

Write the second function **primeDivisors**

```
public static void primeDivisors(int N)
```

which takes input an integer N and prints the prime divisors of N and their largest powers which divide N. You must use the isPrime and largestPower functions given above in **primeDivisors**.

```
N = 120
2 3
3 1
5 1

N = 5040
2 4
3 2
5 1
7 1
```